

Ahsanullah University of Science and Technology (AUST)
Department of Computer Science and Engineering (CSE)

CSE 4214: Pattern Recognition Lab

Experiment No 3

“Designing a Minimum Error Rate classifier”

A. Sessional Task:

Design a Minimum Error Rate Classifier for a two-class problem with the given data below:

$$P(x|\omega_1) = N(\mu_1, \Sigma_1)$$

$$\text{where, } \mu_1 = [0 \ 0]$$

$$\text{and } \Sigma_1 = \begin{bmatrix} .25 & .3 \\ .3 & 1 \end{bmatrix};$$

$$P(\omega_1) = 0.5$$

$$P(x|\omega_2) = N(\mu_2, \Sigma_2)$$

$$\text{where, } \mu_2 = [2 \ 2]$$

$$\text{and } \Sigma_2 = \begin{bmatrix} .5 & 0 \\ 0 & .5 \end{bmatrix};$$

$$P(\omega_2) = 0.5$$

1. Classify the following unknown samples:
 $(1,1), (1,-1), (4,5), (-2,2.5), (0,2), (2,-3)$
2. Classified samples should have different colored markers according to the assigned class label.
3. Change the parameter values (μ_i, Σ_i) to see different recognition results.
4. Label the regions R_i for each of those two classes.
5. Submit your sessional report with necessary illustrations and analysis within 7 days from the day of assigned task.

Note: Useful MATLAB Code:

```
x1 = -7:.2:7; x2 = -7:.2:7;  
[X1,X2] = meshgrid(x1,x2);
```

```
mu1 = [0 0];  
Sigma1 = [.25 .3; .3 1];  
F1 = mvnpdf([X1(:) X2(:)],mu1,Sigma1);  
F1 = reshape(F1,length(x2),length(x1));  
%surf(x1,x2,F1);  
meshc(X1,X2,F1);  
axis([-7 7 -7 7 -1.0 .6])
```

```
xlabel('x1'); ylabel('x2'); zlabel('Probability Density');
```

```
hold on;
```

```
mu2 = [2 2];
```

```
Sigma2 = [.5 .0; 0 .5];
```

```
F2 = mvnpdf([X1(:) X2(:)],mu2,Sigma2);
```

```
F2 = reshape(F2,length(x2),length(x1));
```

```
%surf(x1,x2,F2);
```

```
meshc(X1,X2,F2);
```

```
axis([-7 7 -7 7 -1.0 .6])
```

```
xlabel('x1'); ylabel('x2'); zlabel('Probability Density');
```

```
caxis([min(F2(:))-.5*range(F2(:)),max(F2(:))]);
```

```
%% Write Your CODE here
```

```
plot3(1,1,-1.0,'rx');
```